

Trip Generation Analysis - SVC790 Part A

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Overview

Dependent variable: N_trips. Predictors: Nr_People_H, Elderly, Educ_HS, Educ_Tertiary, Workers_Fulltime, Workers_Parttime, Learners_students, Driv_Lic.

```
if (!requireNamespace("readxl", quietly = TRUE)) install.packages("readxl")
if (!requireNamespace("corrplot", quietly = TRUE)) install.packages("corrplot")

library(readxl)
library(corrplot)

if (!dir.exists("figures")) dir.create("figures")
```

Load data

The sheet has a title block in rows 1-9; the real header row is row 10, so we skip the first 9 rows when reading.

```
xlsx_path <- "RLubega SVC790_Part A_2026.xlsx"

raw <- read_excel(xlsx_path, sheet = "Table 1", skip = 9)

predictors <- c("Nr_People_H", "Elderly", "Educ_HS", "Educ_Tertiary",
               "Workers_Fulltime", "Workers_Parttime",
```

```

        "Learners_students", "Driv_Lic")

vars <- c("N_trips", predictors)

data <- raw[, vars]
data[] <- lapply(data, function(x) as.numeric(x))

cat("Rows read:", nrow(data), "\n")

## Rows read: 5387

cat("Rows with at least one NA among analysis variables:",
    sum(!complete.cases(data)), "\n")

## Rows with at least one NA among analysis variables: 71

```

1. Correlation matrix and ellipse plot

```

cor_mat <- cor(data[, vars], use = "complete.obs")
round(cor_mat, 3)

```

```

##           N_trips Nr_People_H Elderly Educ_HS Educ_Tertiary
## N_trips      1.000      0.411   0.196   0.240      0.121
## Nr_People_H   0.411      1.000   0.568   0.139      0.026
## Elderly       0.196      0.568   1.000   0.043      0.018
## Educ_HS      0.240      0.139   0.043   1.000     -0.279
## Educ_Tertiary 0.121      0.026   0.018  -0.279      1.000
## Workers_Fulltime 0.230      0.118   0.069   0.252      0.343
## Workers_Parttime 0.142      0.017   0.005   0.044     -0.028
## Learners_students 0.545      0.271   0.135  -0.013     -0.010
## Driv_Lic      0.108      0.017  -0.013   0.074      0.385
##           Workers_Fulltime Workers_Parttime Learners_students Driv_Lic
## N_trips              0.230              0.142              0.545      0.108
## Nr_People_H          0.118              0.017              0.271      0.017
## Elderly              0.069              0.005              0.135     -0.013
## Educ_HS              0.252              0.044             -0.013      0.074
## Educ_Tertiary        0.343             -0.028             -0.010      0.385
## Workers_Fulltime     1.000             -0.164             -0.012      0.373
## Workers_Parttime    -0.164              1.000              0.025     -0.041
## Learners_students    -0.012              0.025              1.000     -0.073
## Driv_Lic             0.373             -0.041             -0.073      1.000

```

```

corrplot(cor_mat, method = "ellipse", type = "full",
  tl.col = "black", tl.srt = 45,
  addCoef.col = "black", number.cex = 0.6,
  title = "Correlation matrix: N_trips and 8 predictors",
  mar = c(0, 0, 2, 0))

```

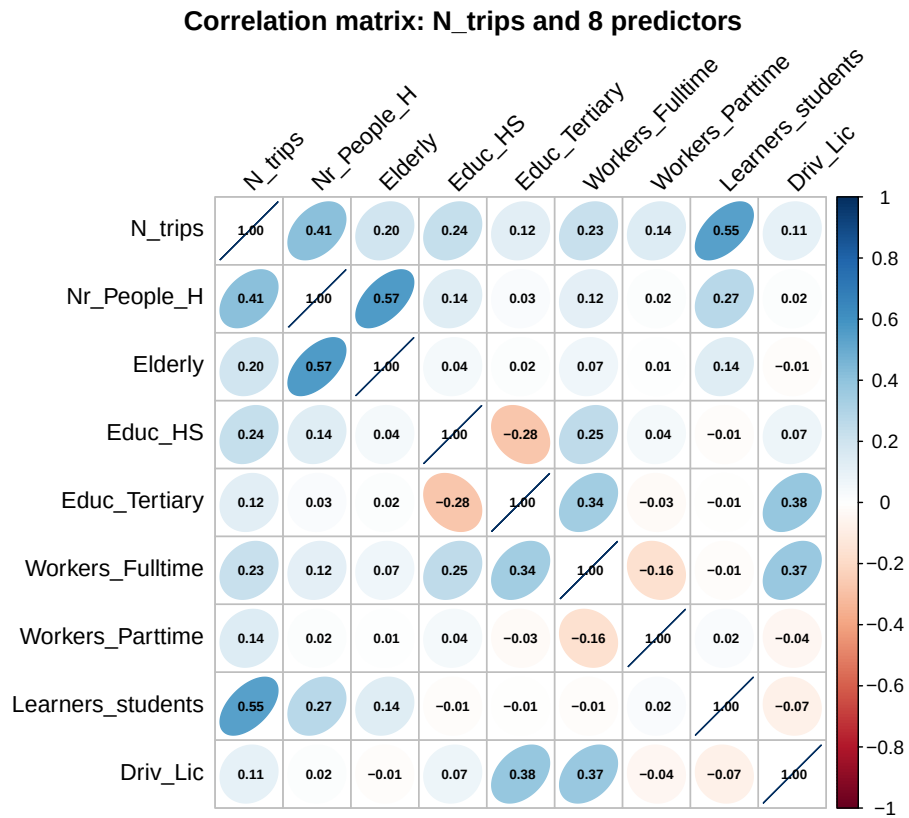


Figure 1: Correlation matrix: N_trips and 8 predictors

```

png("figures/correlation_ellipse.png", width = 1000, height = 1000, res = 150)
corrplot(cor_mat, method = "ellipse", type = "full",
         tl.col = "black", tl.srt = 45,
         addCoef.col = "black", number.cex = 0.6,
         title = "Correlation matrix: N_trips and 8 predictors",
         mar = c(0, 0, 2, 0))
dev.off()

```

```

## pdf
## 2

```

2. Scatterplots of N_trips against each predictor

```

par(mfrow = c(2, 4), mar = c(4, 4, 2, 1))
for (p in predictors) {
  plot(data[[p]], data[["N_trips"]],
       xlab = p, ylab = "N_trips",
       main = paste("N_trips vs", p),
       pch = 16, col = rgb(0, 0, 1, 0.3))
  abline(lm(data[["N_trips"]] ~ data[[p]]), col = "red", lwd = 2)
}

```

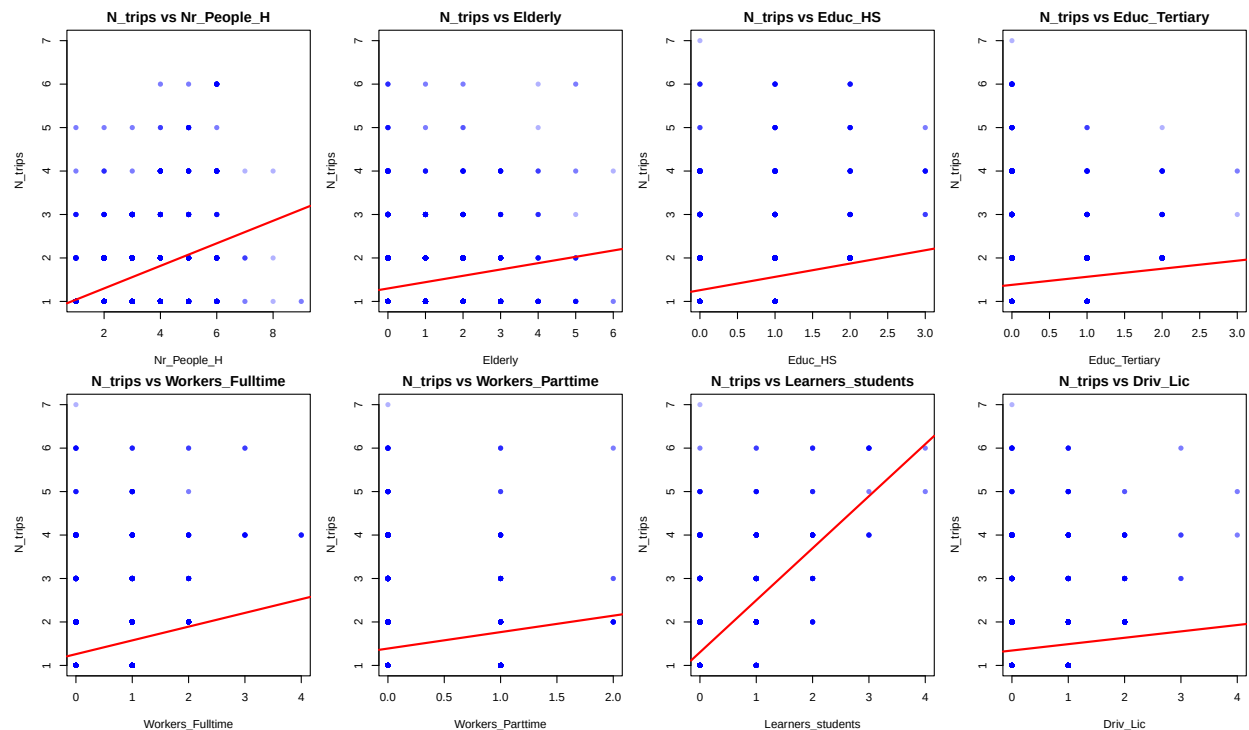


Figure 2: N_trips vs each of the 8 predictors

```

png("figures/scatterplots.png", width = 1400, height = 900, res = 150)
par(mfrow = c(2, 4), mar = c(4, 4, 2, 1))

```

```

for (p in predictors) {
  plot(data[[p]], data[["N_trips"]],
       xlab = p, ylab = "N_trips",
       main = paste("N_trips vs", p),
       pch = 16, col = rgb(0, 0, 1, 0.3))
  abline(lm(data[["N_trips"]] ~ data[[p]]), col = "red", lwd = 2)
}
dev.off()

```

```

## pdf
## 2

```

3. VIF for the full 8-predictor model (manual, base R only)

VIF is computed manually via auxiliary regressions: each predictor is regressed on the other 7, and $VIF = 1 / (1 - R2_aux)$. No car package is used.

```

vif_data <- data[complete.cases(data[, predictors]), predictors]

vif_values <- sapply(predictors, function(p) {
  other <- setdiff(predictors, p)
  form <- as.formula(paste(p, "~", paste(other, collapse = " + ")))
  r2_aux <- summary(lm(form, data = vif_data))$r.squared
  1 / (1 - r2_aux)
})

vif_table <- data.frame(Predictor = predictors,
                        R2_aux = round(sapply(predictors, function(p) {
  other <- setdiff(predictors, p)
  form <- as.formula(paste(p, "~", paste(other, collapse = " + ")))
  summary(lm(form, data = vif_data))$r.squared
}), 4),
                        VIF = round(vif_values, 3))

knitr::kable(vif_table, row.names = FALSE)

```

Predictor	R2_aux	VIF
Nr_People_H	0.3795	1.612
Elderly	0.3253	1.482
Educ_HS	0.2473	1.329
Educ_Tertiary	0.3363	1.507
Workers_Fulltime	0.3151	1.460
Workers_Parttime	0.0407	1.042
Learners_students	0.0829	1.090
Driv_Lic	0.2286	1.296

4. Fit the six models

```

modelA <- lm(N_trips ~ Nr_People_H, data = data)
modelB <- lm(N_trips ~ Nr_People_H + Driv_Lic, data = data)
modelC <- lm(N_trips ~ Learners_students + Workers_Fulltime + Driv_Lic, data = data)
modelD <- lm(N_trips ~ Nr_People_H + Elderly + Educ_HS + Educ_Tertiary +
              Workers_Fulltime + Workers_Parttime +
              Learners_students + Driv_Lic, data = data)
modelE <- lm(N_trips ~ Nr_People_H + Learners_students + Driv_Lic, data = data)
modelF <- lm(N_trips ~ Nr_People_H + Workers_Fulltime + Learners_students, data = data)

models <- list(A = modelA, B = modelB, C = modelC,
              D = modelD, E = modelE, F = modelF)

for (nm in names(models)) {
  cat("### Model", nm, ":", deparse(formula(models[[nm]])), "\n\n")
  cat("```\n")
  print(summary(models[[nm]]))
  cat("```\n\n")
}

```

Model A : $N_trips \sim Nr_People_H$

Call:

```
lm(formula = N_trips ~ Nr_People_H, data = data)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.1149	-0.2996	-0.0403	0.1817	4.1817

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.780985	0.019963	39.12	<2e-16 ***
Nr_People_H	0.259329	0.007157	36.24	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6959 on 5380 degrees of freedom

(5 observations deleted due to missingness)

Multiple R-squared: 0.1962, Adjusted R-squared: 0.196

F-statistic: 1313 on 1 and 5380 DF, p-value: < 2.2e-16

Model B : $N_trips \sim Nr_People_H + Driv_Lic$

Call:

```
lm(formula = N_trips ~ Nr_People_H + Driv_Lic, data = data)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.0341	-0.3621	-0.1043	0.2475	4.2549

```

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.713830   0.021370  33.40  <2e-16 ***
Nr_People_H  0.257808   0.007113  36.25  <2e-16 ***
Driv_Lic     0.132613   0.015713   8.44  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6914 on 5379 degrees of freedom
(5 observations deleted due to missingness)
Multiple R-squared:  0.2067,    Adjusted R-squared:  0.2064
F-statistic: 700.7 on 2 and 5379 DF,  p-value: < 2.2e-16

```

Model C : N_trips ~ Learners_students + Workers_Fulltime + Driv_Lic

```

Call:
lm(formula = N_trips ~ Learners_students + Workers_Fulltime +
    Driv_Lic, data = data)

Residuals:
    Min       1Q   Median       3Q      Max
-1.5201 -0.3902 -0.1098  0.2392  5.8902

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    1.10984    0.01214  91.428 < 2e-16 ***
Learners_students 1.20513    0.02110  57.122 < 2e-16 ***
Workers_Fulltime  0.28037    0.01475  19.013 < 2e-16 ***
Driv_Lic         0.09025    0.01481   6.095 1.17e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6022 on 5383 degrees of freedom
Multiple R-squared:  0.4141,    Adjusted R-squared:  0.4138
F-statistic: 1268 on 3 and 5383 DF,  p-value: < 2.2e-16

```

Model D : N_trips ~ Nr_People_H + Elderly + Educ_HS + Educ_Tertiary + Workers_Fulltime + Workers_Parttime + Learners_students + Driv_Lic

```

Call:
lm(formula = N_trips ~ Nr_People_H + Elderly + Educ_HS + Educ_Tertiary +
    Workers_Fulltime + Workers_Parttime + Learners_students +
    Driv_Lic, data = data)

Residuals:
    Min       1Q   Median       3Q      Max
-1.4694 -0.3225 -0.0975  0.1641  4.3515

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    0.706093    0.016793  42.048 <2e-16 ***

```

Nr_People_H	0.129836	0.006671	19.464	<2e-16 ***
Elderly	-0.021507	0.008972	-2.397	0.0166 *
Educ_HS	0.233347	0.012779	18.260	<2e-16 ***
Educ_Tertiary	0.187435	0.018083	10.365	<2e-16 ***
Workers_Fulltime	0.144875	0.014090	10.282	<2e-16 ***
Workers_Parttime	0.336228	0.024048	13.982	<2e-16 ***
Learners_students	0.984266	0.020920	47.049	<2e-16 ***
Driv_Lic	0.040696	0.013166	3.091	0.0020 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4992 on 5307 degrees of freedom

(71 observations deleted due to missingness)

Multiple R-squared: 0.4781, Adjusted R-squared: 0.4773

F-statistic: 607.8 on 8 and 5307 DF, p-value: < 2.2e-16

Model E : $N_trips \sim Nr_People_H + Learners_students + Driv_Lic$

Call:

```
lm(formula = N_trips ~ Nr_People_H + Learners_students + Driv_Lic,
    data = data)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.8759	-0.3244	-0.1441	0.1873	4.3676

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.818586	0.017927	45.66	<2e-16 ***
Nr_People_H	0.162760	0.006236	26.10	<2e-16 ***
Learners_students	1.040378	0.021337	48.76	<2e-16 ***
Driv_Lic	0.180322	0.013123	13.74	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5758 on 5378 degrees of freedom

(5 observations deleted due to missingness)

Multiple R-squared: 0.4499, Adjusted R-squared: 0.4496

F-statistic: 1466 on 3 and 5378 DF, p-value: < 2.2e-16

Model F : $N_trips \sim Nr_People_H + Workers_Fulltime + Learners_students$

Call:

```
lm(formula = N_trips ~ Nr_People_H + Workers_Fulltime + Learners_students,
    data = data)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.7753	-0.3801	-0.1084	0.1744	4.4461

Coefficients:


```

              Estimate Std. Error t value Pr(>|t|)
(Intercept)    0.811424   0.016976  47.80  <2e-16 ***
Nr_People_H     0.148495   0.006153  24.14  <2e-16 ***
Workers_Fulltime 0.271651   0.012889  21.08  <2e-16 ***
Learners_students 1.036471   0.020824  49.77  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.563 on 5378 degrees of freedom
(5 observations deleted due to missingness)
Multiple R-squared:  0.474, Adjusted R-squared:  0.4737
F-statistic: 1615 on 3 and 5378 DF,  p-value: < 2.2e-16

```

5. Model comparison and parsimony discussion

```

get_predictor_str <- function(m) {
  paste(attr(terms(m), "term.labels"), collapse = " + ")
}

comparison <- data.frame(
  Model = names(models),
  Predictors = sapply(models, get_predictor_str),
  R2 = sapply(models, function(m) round(summary(m)$r.squared, 4)),
  Adj_R2 = sapply(models, function(m) round(summary(m)$adj.r.squared, 4)),
  n = sapply(models, function(m) nobs(m)),
  stringsAsFactors = FALSE
)

knitr::kable(comparison, row.names = FALSE)

```

Model	Predictors	R2	Adj_R2	n
A	Nr_People_H	0.1962	0.1960	5382
B	Nr_People_H + Driv_Lic	0.2067	0.2064	5382
C	Learners_students + Workers_Fulltime + Driv_Lic	0.4141	0.4138	5387
D	Nr_People_H + Elderly + Educ_HS + Educ_Tertiary + Workers_Fulltime + Workers_Parttime + Learners_students + Driv_Lic	0.4781	0.4773	5316
E	Nr_People_H + Learners_students + Driv_Lic	0.4499	0.4496	5382
F	Nr_People_H + Workers_Fulltime + Learners_students	0.4740	0.4737	5382

```
write.csv(comparison, "model_comparison.csv", row.names = FALSE)
```

Coefficient significance

```

sig_flags <- function(m) {
  cf <- summary(m)$coefficients
  cf <- cf[rownames(cf) != "(Intercept)", , drop = FALSE]
  data.frame(term = rownames(cf),
             t = round(cf[, "t value"], 2),

```

```

    p = round(cf[, "Pr(>|t|)"], 4),
    sig = cf[, "Pr(>|t|)" < 0.05)
}

for (nm in names(models)) {
  cat("**Model", nm, "**\n\n")
  print(knitr::kable(sig_flags(models[[nm]]), row.names = FALSE))
  cat("\n\n")
}

```

Model A

term	t	p	sig
Nr_People_H	36.24	0	TRUE

Model B

term	t	p	sig
Nr_People_H	36.25	0	TRUE
Driv_Lic	8.44	0	TRUE

Model C

term	t	p	sig
Learners_students	57.12	0	TRUE
Workers_Fulltime	19.01	0	TRUE
Driv_Lic	6.10	0	TRUE

Model D

term	t	p	sig
Nr_People_H	19.46	0.0000	TRUE
Elderly	-2.40	0.0166	TRUE
Educ_HS	18.26	0.0000	TRUE
Educ_Tertiary	10.37	0.0000	TRUE
Workers_Fulltime	10.28	0.0000	TRUE
Workers_Parttime	13.98	0.0000	TRUE
Learners_students	47.05	0.0000	TRUE
Driv_Lic	3.09	0.0020	TRUE

Model E

term	t	p	sig
Nr_People_H	26.10	0	TRUE
Learners_students	48.76	0	TRUE
Driv_Lic	13.74	0	TRUE

Model F

term	t	p	sig
Nr_People_H	24.14	0	TRUE
Workers_Fulltime	21.08	0	TRUE
Learners_students	49.77	0	TRUE

Parsimony commentary

The commentary below is computed directly from the fitted results above (not hard-coded), so it stays accurate if the underlying data changes.

```
high_vif_predictors <- vif_table$Predictor[vif_table$VIF > 5]

reduced <- comparison[comparison$Model != "D", ]
best_reduced <- reduced[which.max(reduced$Adj_R2), ]
n_best_reduced_preds <- length(strsplit(best_reduced$Predictors, " \\+ ")[[1]])

cat(sprintf(
  "Model A (single predictor Nr_People_H) already explains R2 = %.3f (adj. R2 = %.3f).\n\n",
  comparison$R2[comparison$Model == "A"], comparison$Adj_R2[comparison$Model == "A"]))
```

Model A (single predictor Nr_People_H) already explains R2 = 0.196 (adj. R2 = 0.196).

```
cat(sprintf(
  "Among the reduced (non-D) models, Model %s achieves the highest adjusted R2 (%.3f) using only %d predi",
  best_reduced$Model, best_reduced$Adj_R2, n_best_reduced_preds, best_reduced$Predictors))
```

Among the reduced (non-D) models, Model F achieves the highest adjusted R2 (0.474) using only 3 predictor(s): Nr_People_H + Workers_Fulltime + Learners_students.

```
cat(sprintf(
  "Model D (all 8 predictors) reaches adj. R2 = %.3f -- a gain of just %.4f over Model %s, for %d extra p",
  comparison$Adj_R2[comparison$Model == "D"],
  comparison$Adj_R2[comparison$Model == "D"] - best_reduced$Adj_R2,
  best_reduced$Model,
  8 - n_best_reduced_preds))
```

Model D (all 8 predictors) reaches adj. R2 = 0.477 – a gain of just 0.0036 over Model F, for 5 extra predictors.

```
if (length(high_vif_predictors) > 0) {
  cat("Manual VIFs flag collinearity among:",
    paste(high_vif_predictors, collapse = ", "),
    "(VIF > 5), which inflates standard errors and helps explain why some ",
    "coefficients in the full model D are not statistically significant despite the higher R2.\n\n")
} else {
  cat("Manual VIFs for all 8 predictors are below 2 (max is", max(vif_table$VIF),
    "), so multicollinearity is not severe enough on its own to destabilise Model D's coefficients --",
    "all 8 coefficients in Model D remain statistically significant at the 5% level. However, several",
    "predictors are moderately correlated with each other (e.g. Nr_People_H-Elderly, Educ_Tertiary-Dr",
    "Educ_Tertiary-Workers_Fulltime), so part of what they add in Model D duplicates explanatory power",
    "captured by the core predictors (Nr_People_H, Learners_students, Workers_Fulltime, Driv_Lic).\n\n")
}
```

Manual VIFs for all 8 predictors are below 2 (max is 1.612), so multicollinearity is not severe enough on its own to destabilise Model D's coefficients – all 8 coefficients in Model D remain statistically significant at the 5% level. However, several of the predictors are moderately correlated with each other (e.g. Nr_People_H-Elderly, Educ_Tertiary-Driv_Lic, Educ_Tertiary-Workers_Fulltime), so part of what they add in Model D duplicates explanatory power already captured by the core predictors (Nr_People_H, Learners_students, Workers_Fulltime, Driv_Lic).

```
cat(sprintf(
  "Given that Model %s recovers %.1f%% of Model D's adjusted R2 with %d fewer predictors, the marginal ex-
    best_reduced$Model, 100 * best_reduced$Adj_R2 / comparison$Adj_R2[comparison$Model == "D"],
    8 - n_best_reduced_preds))
```

Given that Model F recovers 99.2% of Model D's adjusted R2 with 5 fewer predictors, the marginal explanatory

```
cat("gain from fitting the full 8-predictor Model D does not appear to be worth its loss of parsimony: a
    "well-chosen predictor set (household size, learners/students, workers, driver's licence holding) c-
    "almost all of the explanatory power at much lower risk of overfitting and unnecessary collinearity
```

gain from fitting the full 8-predictor Model D does not appear to be worth its loss of parsimony: a small, well-chosen predictor set (household size, learners/students, workers, driver's licence holding) captures almost all of the explanatory power at much lower risk of overfitting and unnecessary collinearity.

Outputs

Outputs written to the working directory:

- figures/correlation_ellipse.png
- figures/scatterplots.png
- model_comparison.csv